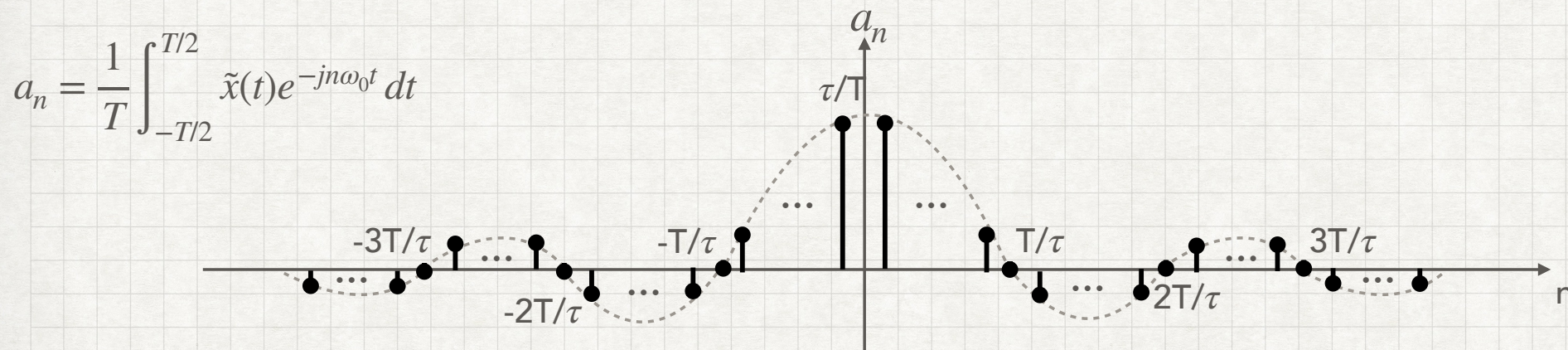


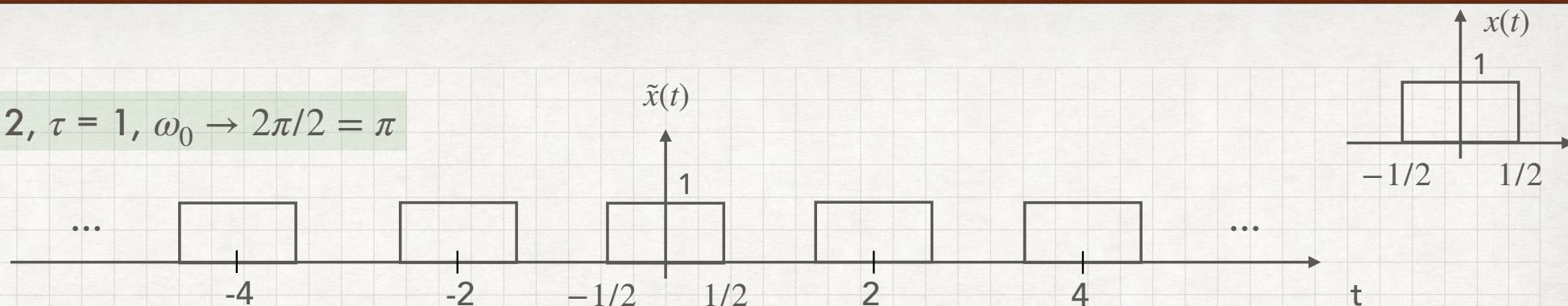
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $\text{rect}(t/\tau)$ WITH PERIOD $T \leftrightarrow \frac{\tau}{T} \text{sinc}\left(\frac{n\tau}{T}\right)$



$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt$$

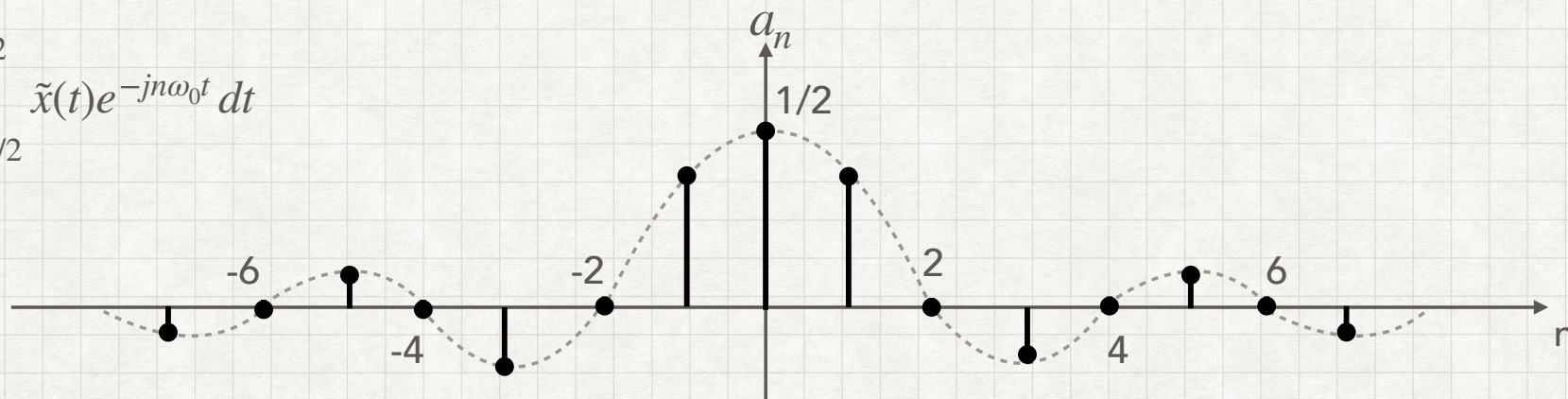
$$T = 2, \tau = 1, \omega_0 \rightarrow 2\pi/2 = \pi$$



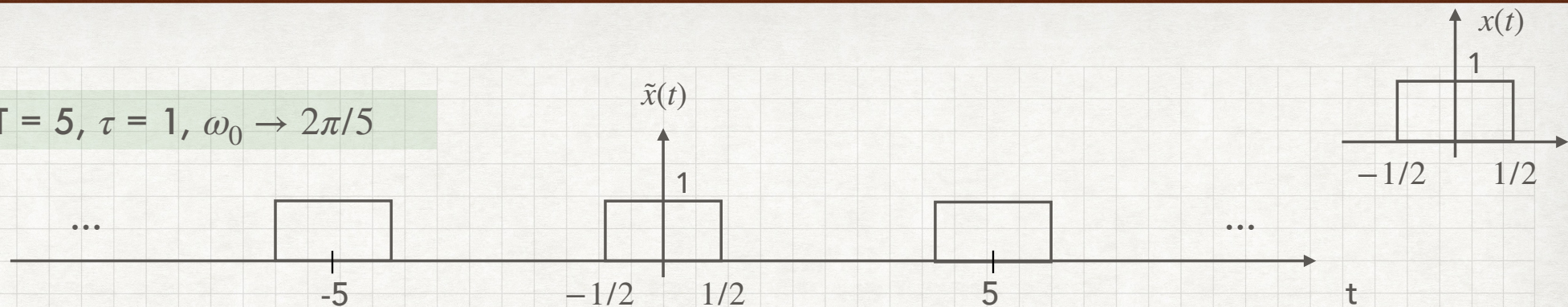
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $rect(t)$ WITH $T=2 \leftrightarrow \frac{1}{2} \text{sinc}\left(\frac{n}{2}\right)$

$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt$$



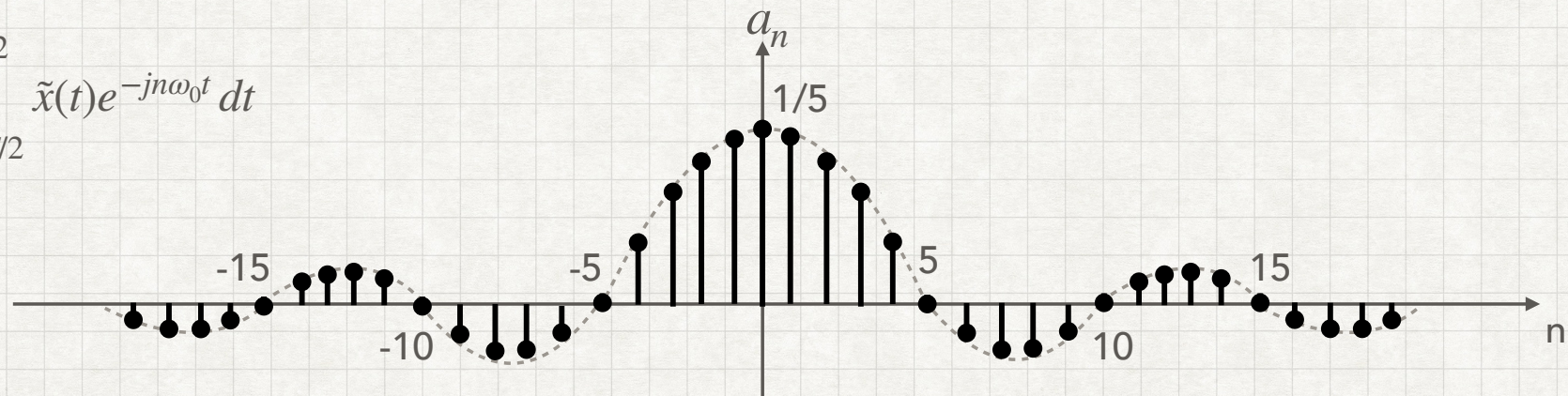
$$T = 5, \tau = 1, \omega_0 \rightarrow 2\pi/5$$



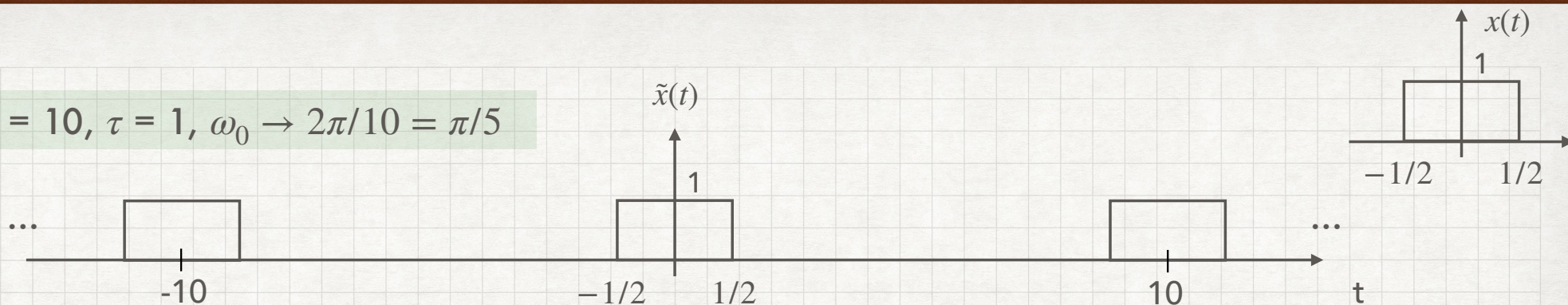
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $rect(t)$ WITH $T=5 \leftrightarrow \frac{1}{5} sinc(\frac{n}{5})$

$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt$$



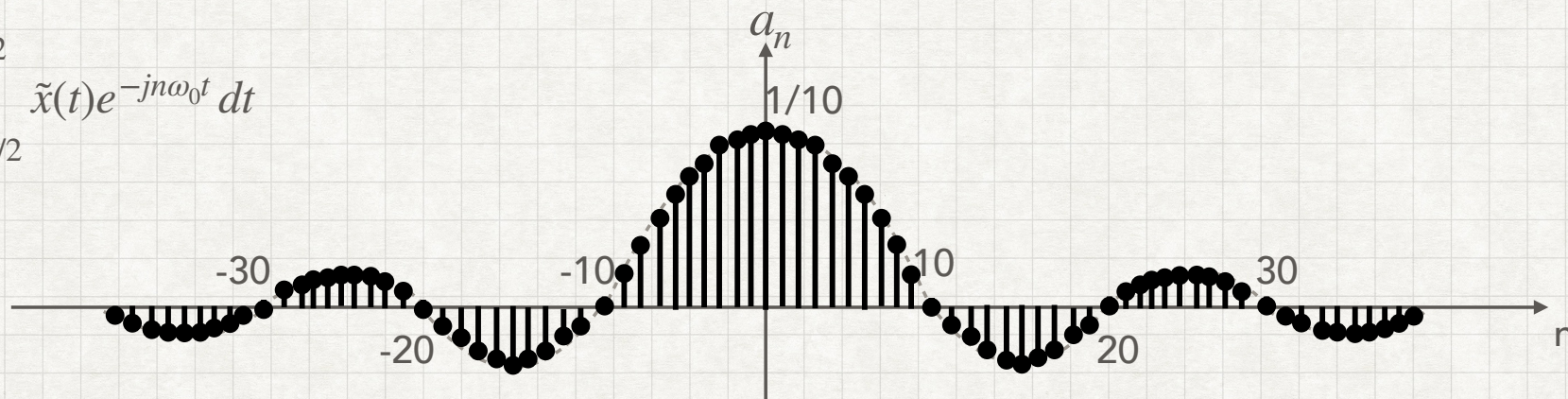
$$T = 10, \tau = 1, \omega_0 \rightarrow 2\pi/10 = \pi/5$$



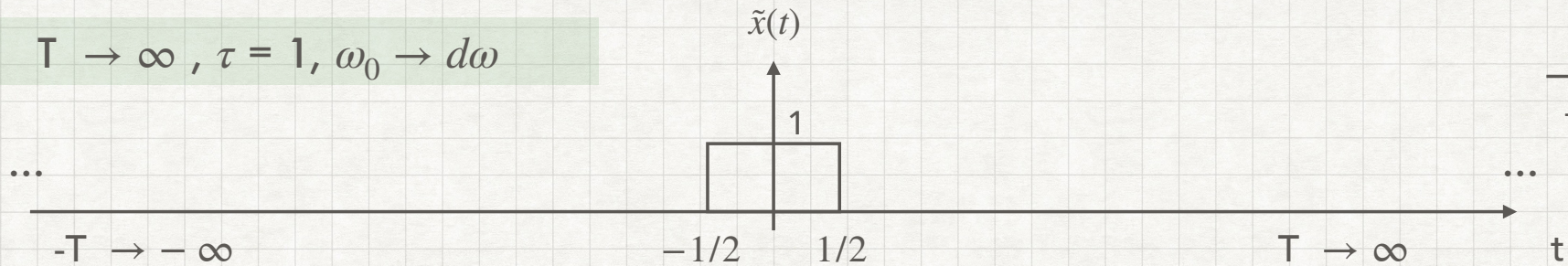
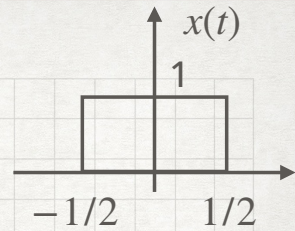
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $rect(t)$ WITH $T=10 \leftrightarrow \frac{1}{10} \text{sinc}\left(\frac{n}{10}\right)$

$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt$$



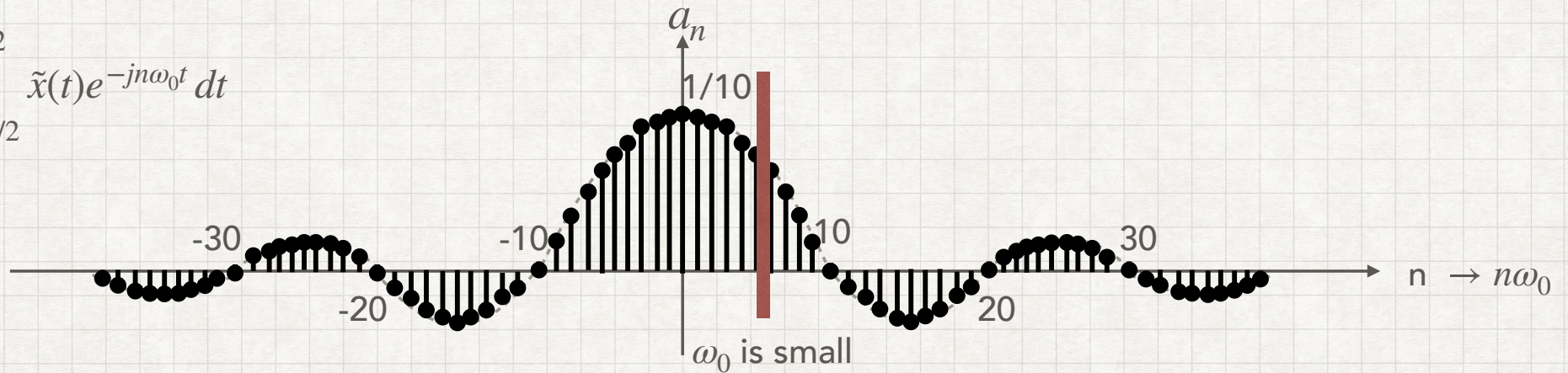
$$T \rightarrow \infty, \tau = 1, \omega_0 \rightarrow d\omega$$



$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

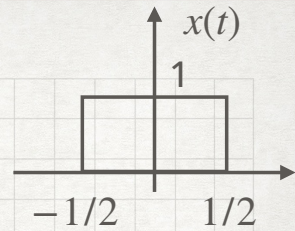
A PERIODIC SIGNAL $\text{rect}(t)$ WITH $T \rightarrow \infty \leftrightarrow$ _____

$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt$$



$$T \rightarrow \infty, \tau = 1, \omega_0 \rightarrow d\omega$$

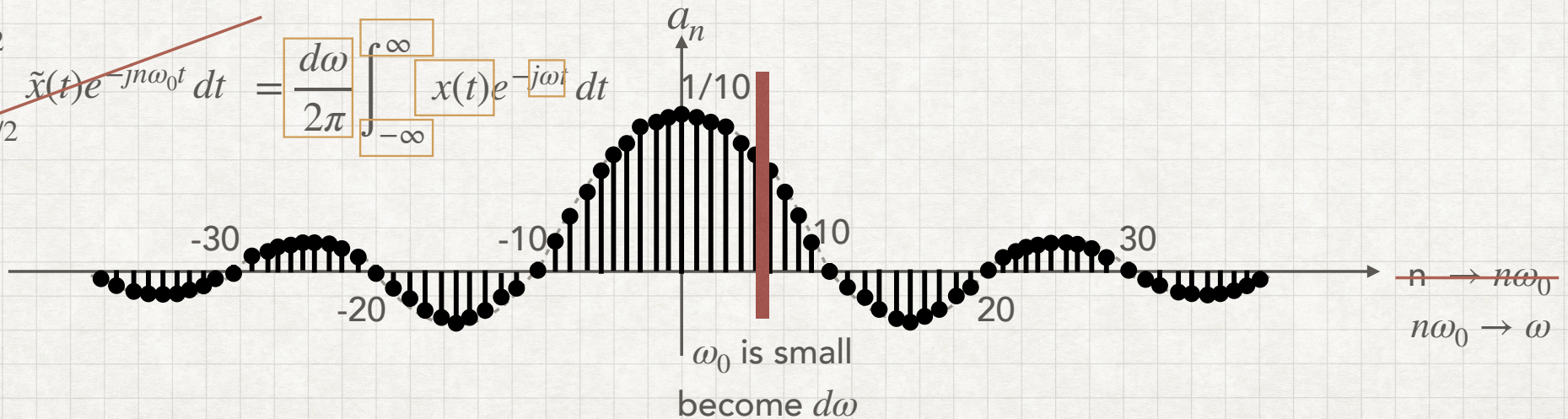
$$\tilde{x}(t) \rightarrow x(t)$$



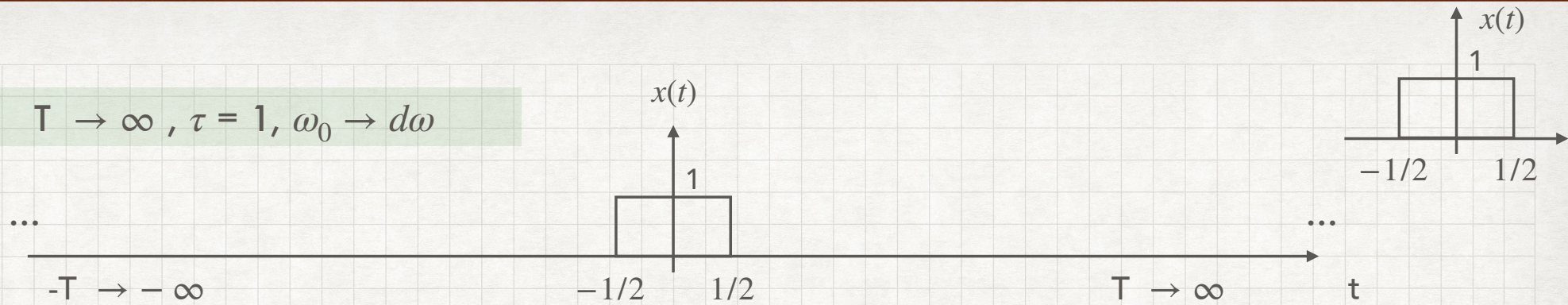
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $rect(t)$ WITH $T \rightarrow \infty \leftrightarrow$ _____

$$a_n = \frac{1}{T} \int_{-T/2}^{T/2} \tilde{x}(t) e^{-jn\omega_0 t} dt = \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$



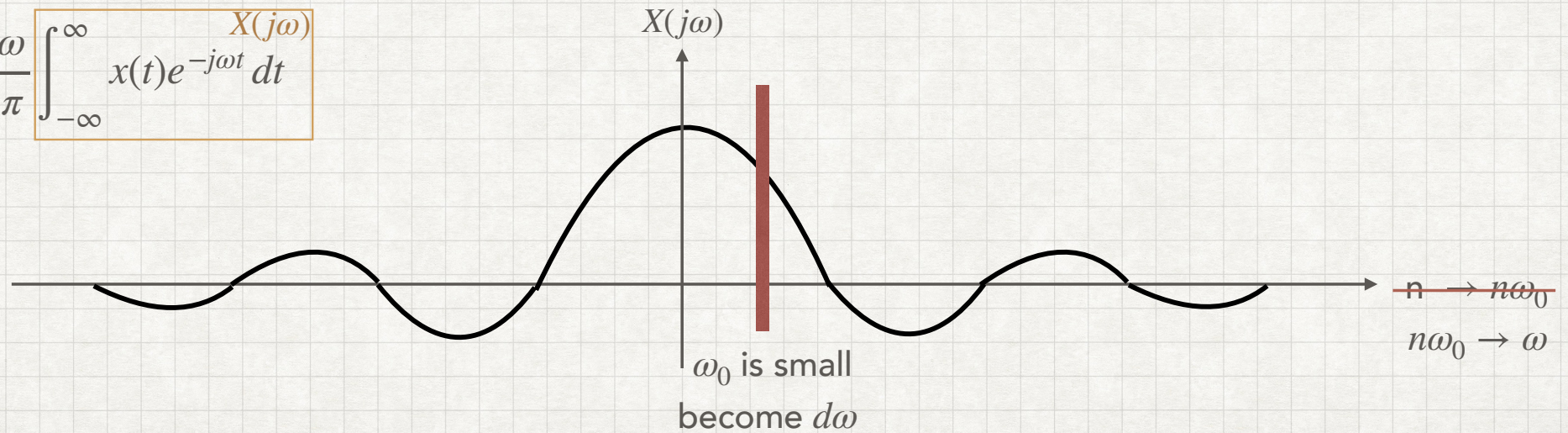
$$T \rightarrow \infty, \tau = 1, \omega_0 \rightarrow d\omega$$



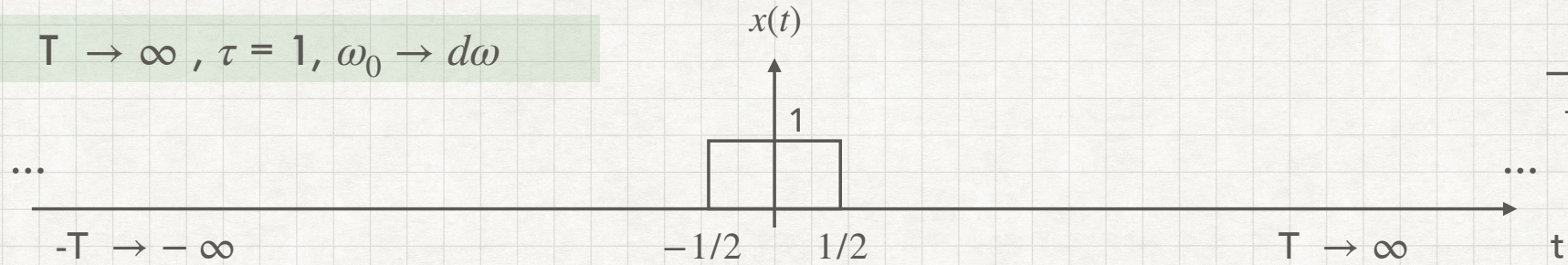
$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $\text{rect}(t)$ WITH $T \rightarrow \infty \leftrightarrow$ _____

$$a_n = \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$



$$T \rightarrow \infty, \tau = 1, \omega_0 \rightarrow d\omega$$



$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} a_n e^{jn(\omega_0)t}$$

A PERIODIC SIGNAL $\text{rect}(t)$ WITH $T \rightarrow \infty \leftrightarrow$ _____

$$a_n = \frac{d\omega}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

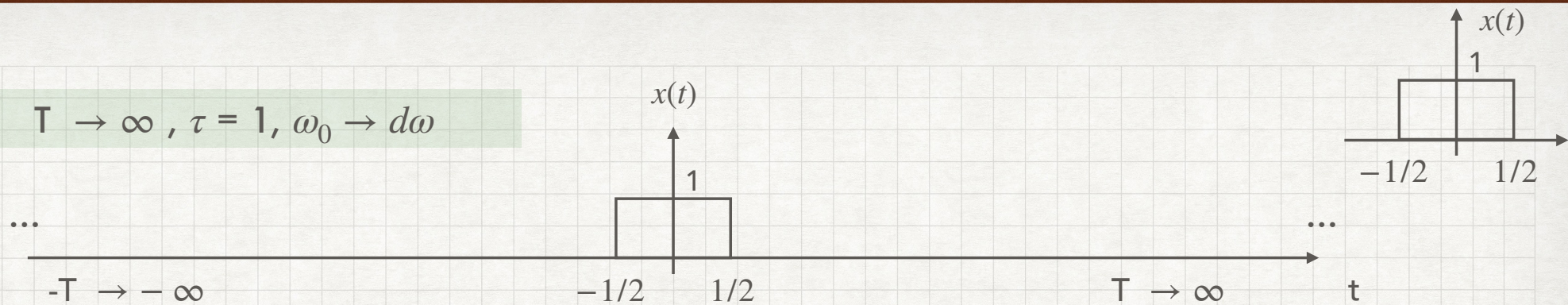
$$x(t) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right] e^{j\omega t} \frac{d\omega}{2\pi}$$

$X(j\omega)$

ω_0 is small
become $d\omega$

$n \rightarrow n\omega_0$
 $n\omega_0 \rightarrow \omega$

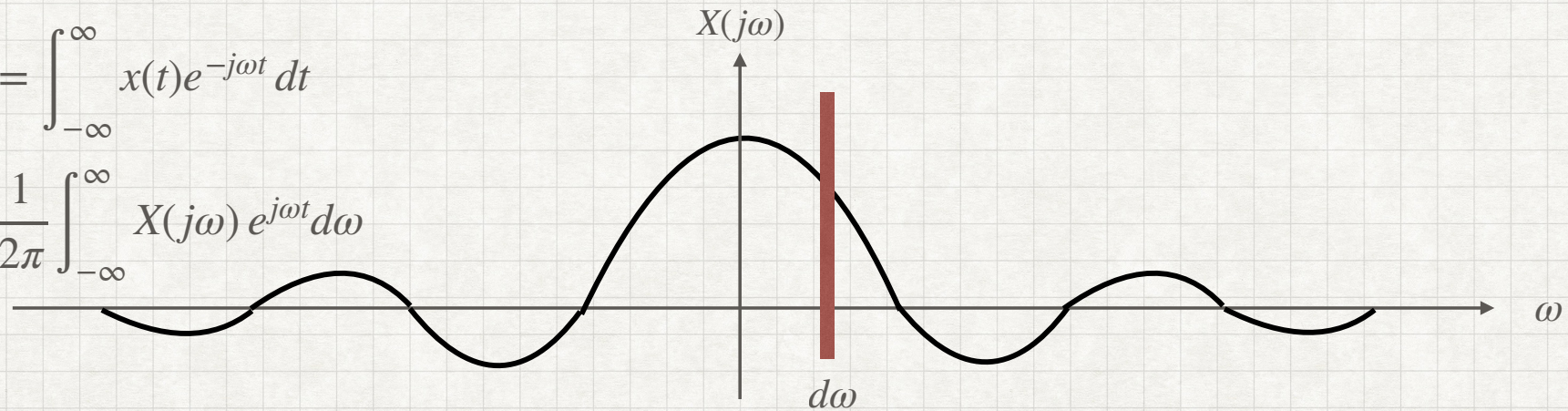
$$T \rightarrow \infty, \tau = 1, \omega_0 \rightarrow d\omega$$



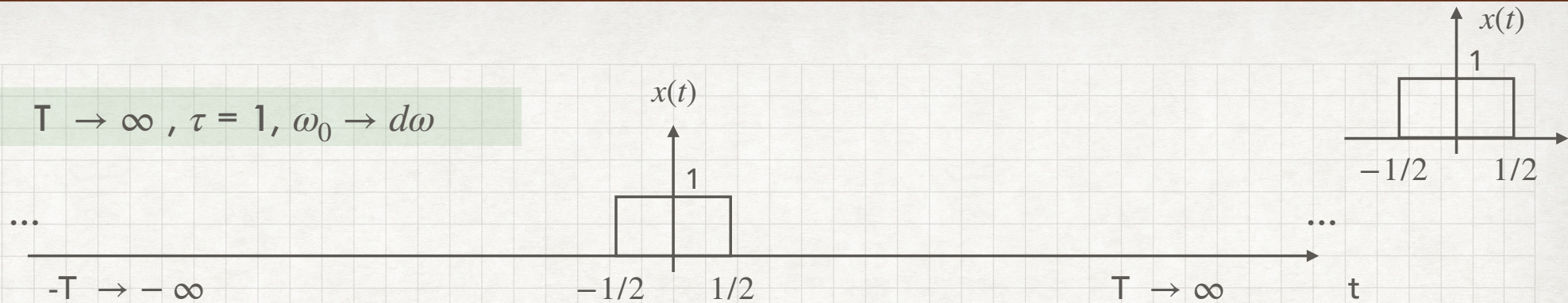
A PERIODIC SIGNAL $\text{rect}(t)$ WITH $T \rightarrow \infty \leftrightarrow$ _____

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$



$$T \rightarrow \infty, \tau = 1, \omega_0 \rightarrow d\omega$$



A PERIODIC SIGNAL $\text{rect}(t)$ WITH $T \rightarrow \infty \leftrightarrow \text{sinc}(\frac{\omega}{2})$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

